

STAT 24410/30030 Final Review Problems

1. **Lecture Notes - Further Exercises** (at the end of each chapter).

2. The Poisson (λ) distribution ($\lambda > 0$) has the following probability mass function:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$$

with mean and variance equal to λ .

(a) Let $X_1, \dots, X_n$ be iid Poisson (λ). Find the MLE of λ .

(b) For testing $H_0 : \lambda = 3$ versus $H_A : \lambda \neq 3$, find the generalized likelihood ratio Λ and its asymptotic distribution.

Note: The Gamma (α, β) distribution ($\alpha, \beta > 0$) has density:

$$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0$$

with mean α/β and variance α/β^2 .

(c) Suppose now that $X_1, \dots, X_n$ are iid Poisson (λ), and that we have a prior for λ given by a Gamma (α, β) distribution. Find the posterior distribution for λ .

(d) Calculate the Bayes estimator with respect to squared loss when $n=2$, $X_1 = 29$, $X_2 = 13$ and the prior is Gamma (20, 10). Comment on how it compares with the MLE.

3. Suppose that Y_1 and Y_2 are independent Poisson distributed random variables with means λ_1 and λ_2 , respectively. Let $W = Y_1 + Y_2$. Use the fact that W has a Poisson distribution with mean $\lambda_1 + \lambda_2$ to show that the conditional distribution of Y_1 , given that $W = w$, is a binomial distribution with $n = w$ and $p = \frac{\lambda_1}{\lambda_1 + \lambda_2}$.

4. You draw one data point X and would like to test which distribution it came from. There are two possibilities:

- H_0 : X was drawn from a Uniform[0, 1] distribution.
- H_1 : X was drawn from the distribution with density $f(x) = 1 - 0.5x$ supported on $x \in [0, 2]$.

(a) Suppose you decide to perform a likelihood ratio test. Write an equation for the likelihood ratio LR as a function of the observed value $X = x$.

(b) What is the region of X values for which you would choose H_0 , and what is the region for which

you would choose H_1 , if you set your likelihood ratio threshold at $c = 1.5$?

(c) What is the Type I error of this test?

(d) What is the power of this test?

5. Consider the following Bayesian model: a single data point X is drawn from a $\text{Unif}[0, \theta]$ distribution, where θ is an unobserved parameter. Prior distribution on θ is $\theta \sim \text{Uniform}(0, 2)$.

(a) Write the prior density $f_\theta(\theta)$ and the data density $f(x|\theta)$.

(b) What is the posterior density of θ after observing a single draw X of the data? You should have the density $f(\theta|x)$, given exactly (i.e. you may need to solve for a normalizing constant), and be sure to specify the support/range of θ values.

(c) What is the MAP (maximum a posteriori, i.e. the mode of posterior) estimator for θ , after observing one data point $X = x$?

6. Prove that under square error loss the Bayes estimator is the posterior mean, and that under absolute error loss the Bayes estimator is the median of the posterior distribution.